Disturbance rejection with a gyro-based reduced-order observer for the LUCY pointer: implementation and results on a tracking task

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Purpose. This report describes, step by step, how the disturbance observer proposed by Hilkert and Pautler [1] was added to the time-domain model of the LUCY pointer, and how it was compared with the current control structure on the tracking task of Attorrese [4]: a drone flying a circle of radius 10 m at 5 m of height, 100 m away, at 0.2 rad/s. The comparison is done twice:

- with a continuous loop and an ideal gyro, to see what the structure can do;
- with the constraints of the real signals: drives on an EtherCAT network with process data every 1 ms, and a MEMS gyro (Analog Devices ADIS16507 [5]) read over SPI, with its second-order dynamics and the noise of its Allan deviation.

The focus is on the pointing error and on the jitter of the line of sight, current structure against the observer with the noisy gyro.

Main results (EtherCAT and ADIS16507, velocity feedforward of the drives on, one orbit; line of sight of the telescope).

- **Pointing error:** 26.6 mdeg rms (131 mdeg max) with the current structure, 2.72 mdeg rms (8.0 mdeg max) with the observer: 10 times lower in rms, 16 times lower at the peak.
- **Jitter above 1 Hz:** 99 μrad with the current structure, 9.7 μrad with the observer and the noisy gyro (5.0 μrad with an ideal one): 10 times lower.
- **Jitter above 10 Hz:** slightly higher with the observer (2.61 against 1.88 μrad), and much higher above 100 Hz (0.14 μrad against $4.6{\cdot}10^{-4}$ μrad, negligible in absolute terms). This is the gyro noise, differentiated by the observer and fed to the motor (Hilkert's noise-coupling trade-off [1, eq. 9]).
- **Without the velocity feedforward** (as in the tests of October) the pointing error is the following error of the position loop, about 124 mdeg rms, and the observer cannot reduce it: by construction it does not change the response to the command.
- **Required torque:** with an ideal gyro the observer asks for the same torque as the current structure (rms 0.65 and 0.63 N m in azimuth), only earlier at the velocity reversals. With the ADIS16507 the azimuth torque carries 0.33 N m rms of noise above 10 Hz, with a peak of 2.69 N m (40 % of the peak torque of the motor).
- EtherCAT at 1 ms changes the pointing error by hundredths of a millidegree; the observer must pass through a notch on the first structural mode of the elevation to be stable above 7 Hz.

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1 Task and model

1.1 Tracking task

The reference is the one of Attorrese [4, sec. 6.2] (Figure 1): pointer at the origin, drone on a horizontal circle of radius $R = 10\,\mathrm{m}$ at height $h = 5\,\mathrm{m}$, centred at $(100, 0)\,\mathrm{m}$, angular rate $\omega = 0.2\,\mathrm{rad/s}$ (period 31.4 s). With the drone at (x, y, h), the pointing angles and rates are

$$\psi = \operatorname{atan2}(y, x), \quad \theta = \operatorname{atan2}\big(h, \rho\big), \quad \rho = \sqrt{x^2 + y^2}, \quad \dot{\psi} = \frac{x\dot{y} - y\dot{x}}{\rho^2}, \quad \dot{\theta} = -\frac{h\,\dot{\rho}}{\rho^2 + h^2}. \tag{1.1}$$

- The azimuth swings $\pm 5.74°$ at up to 1.27 deg/s, with two reversals per orbit.
- The elevation moves between 2.60 and 3.18° at up to 0.059 deg/s: always in the low-speed range of the friction.

The pointer starts at rest on the reference; the metrics use one orbit after 5 s of settling.

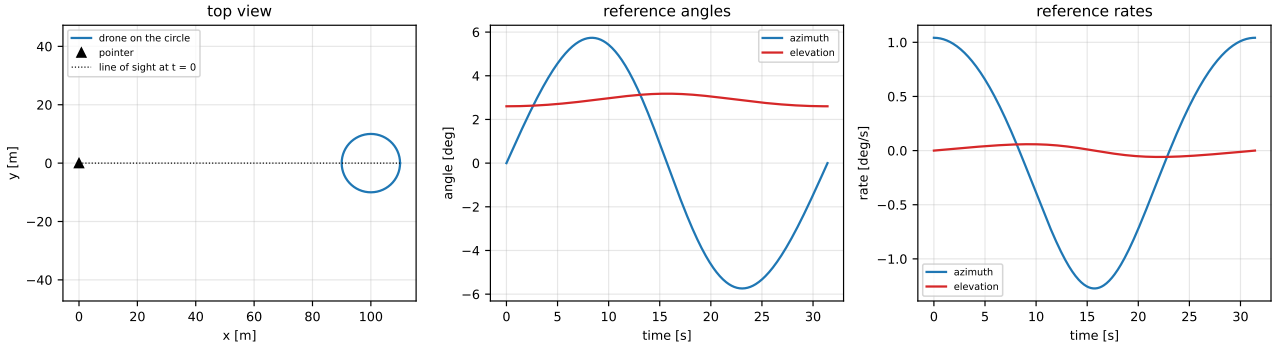


Figure 1: Reference: top view of the circle [m], pointing angles [deg] and rates [deg/s] against time [s].

1.2 Model of the pointer

The time-domain model is the one of the earlier reports: the 8-DOF lumped plant H6 (plant fitted on the October frequency responses), the TBM7646 motors and the AKD2G drives, with

- the effective gains of the drives identified on the step tests (iteration I5);
- the bearing friction (Dahl element with a Stribeck level), the cable spring $K_c(\theta - \theta_0)$ and the torque scale κ_i identified on the V2 doublets.

Table 1: Parameters of the model per axis: drive gains (I5), torque scale, friction (Coulomb level, break-away level, Stribeck speed, pre-sliding stiffness), cable spring (V2), and the rigid inertia $J' = \mathbf{r}^T\mathbf{M}\mathbf{r}$ used by the observer.

axis	K_{pp} [1/s]	K_{pv} [A s/rad]	K_{iv} [Hz]	κ_i [–]	T_c [N m]	T_s [N m]	v_s [deg/s]	σ_0 [N m/rad]	K_c [N m/rad]	θ_0 [deg]	J' [kg m^2]
azimuth	6.98	379	0.26	0.96	0.23	1.84	0.33	4513	0.93	-2.68	2.162
elevation	6.78	54	2.88	1.07	0.41	0.46	0.86	1703	1.66	-2.16	0.900

2 Control architectures

2.1 Current structure

In the terms of Hilkert's review of inertially stabilised platforms [2], LUCY today is a gimbal pointed by encoder (position) loops, with no inertial sensor. Each drive closes a position loop (P) and a velocity loop (PI) on the encoder of its axis, with the forward biquads (notches) on the velocity error, the feedback biquads and the velocity observer of the drive on the velocity feedback, and an inner current loop (Figure 2). The velocity feedforward K_{vff} exists in the drive files ($K_{vff} = 1$) but was not active in the tests; both settings are studied. The line of sight (LOS) is the telescope, which the encoder does not see: in azimuth the encoder sits on the motor side of the shaft that carries the fork.

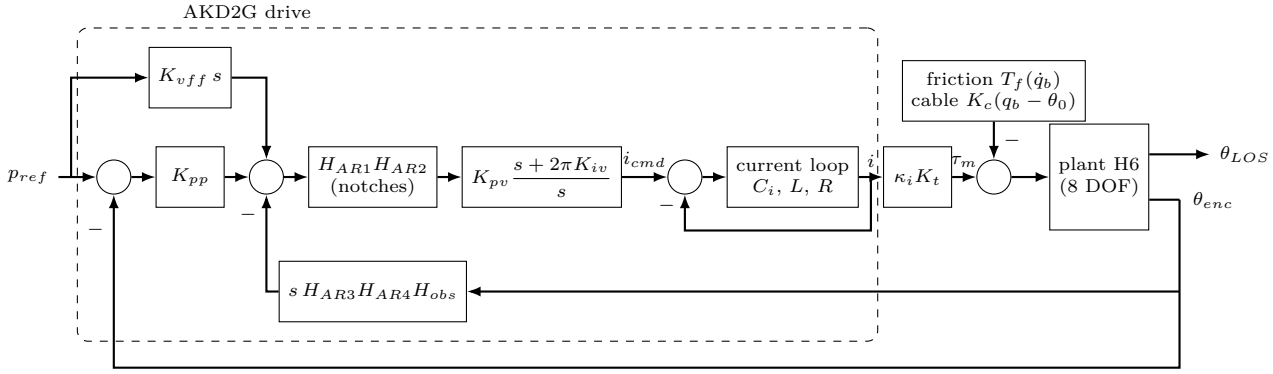


Figure 2: Current structure, one axis: position P and velocity PI of the drive closed on the encoder (θ_{enc}), velocity feedforward K_{vff} (off in the tests), forward notches on the velocity error, current loop, motor torque $\kappa_i K_t i$ on the plant, which is also loaded by the bearing friction and the cable spring. The pointing error is measured on the line of sight θ_{LOS} (telescope).

2.2 Proposed structure: gyro and reduced-order disturbance observer (Hilkert)

Hilkert and Pautler [1] add to an inertially stabilised loop a *single-state* disturbance observer: a model of the rigid gimbal driven by the measured motor torque, closed on the gyro rate by one gain K_o (Figure 3):

$$\dot{\hat{\omega}} = \frac{K_t' i_m + \hat{T}_d}{J'}, \qquad \hat{T}_d = K_o\,(\omega_g - \hat{\omega}) \qquad \Longleftrightarrow \qquad \hat{T}_d = \underbrace{\frac{K_o/J'}{s + K_o/J'}}_{\text{filter}} \underbrace{\left(J' s\,\omega_g - K_t' i_m\right)}_{\text{disturbance measurement}}, \tag{2.1}$$

which is a filtered measurement of the disturbance from Newton's law $T_m + T_d = J\dot{\omega}$ [1, eqs. 1–2]. The estimate is subtracted from the current command, $i_{ff} = -\hat{T}_d/K_t'$, so that the motor cancels the disturbance. In the ideal case (rigid gimbal, exact parameters) the observer leaves the command response unchanged and multiplies the disturbance rejection by $s/(s + K_o/J')$ [1, eqs. 5–6]:

$$\frac{\omega_o}{T_d/J} = \frac{1}{s + G_c/J} \cdot \frac{s}{s + K_o/J}. \tag{2.2}$$

For LUCY (Figure 4) the loops of the drives stay as they are; a rate gyro is added on the telescope and the observer output enters the current command, the point where Hilkert subtracts it when no notch filter sits before the amplifier. As in Hilkert's figure 3, the estimate can pass through a notch on the structural mode.

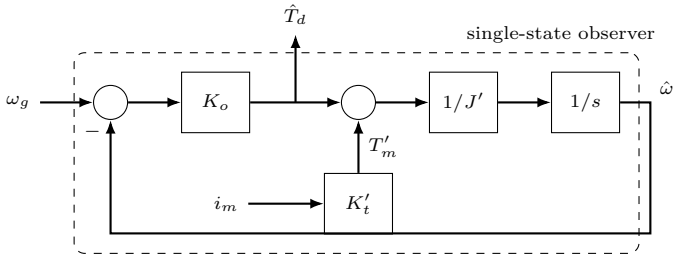


Figure 3: Single-state disturbance observer of Hilkert and Pautler [1, fig. 3]: the model of the rigid axis ($1/J's$) is driven by the motor torque measured through the current, $T'_m = K'_t i_m$, and closed on the gyro rate by the gain K_o; the correction is the estimate $\hat{T}_d$ of the disturbance.

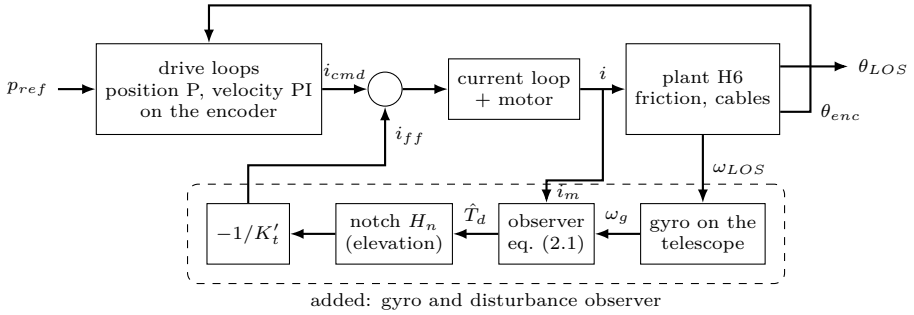


Figure 4: Proposed structure, one axis: the loops of the drive on the encoder are unchanged; a rate gyro on the telescope and the observer of Figure 3 give the disturbance estimate, which (through a notch on the structural mode in elevation) is subtracted from the current command.

2.3 Signal constraints: EtherCAT and the MEMS gyro

In the real system the observer runs in the EtherCAT master, which exchanges the process data with the drives once per cycle, and reads the gyro over SPI (Figure 5).

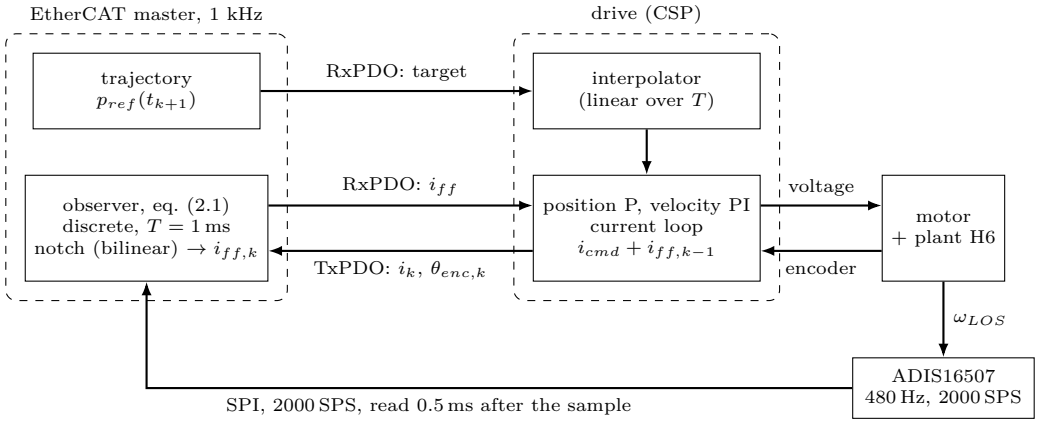


Figure 5: Sampled-data structure: the master sends each cycle the position target (interpolated by the drive) and the feedforward current computed in the previous cycle (RxPDO), reads the actual current and position (TxPDO) and the gyro over SPI; the drive loops run inside the drive.

3 Implementation step by step

3.1 Step 1: closed loop with the identified disturbances

The closed loop of `lucy_cl` (`ClosedLoop`) received two optional attributes, neutral by default so that the earlier studies and the tests are unchanged:

- `torque_scale`: motor torque $\kappa_i K_t i$ instead of $K_t i$;
- `cable_spring`: torque $-K_c(q_b - \theta_0)$ on the bearing node of each axis, added to the generalised forces.

The drives take the I5 gains; the friction elements take the V2 parameters (Table 1). The initial state is at rest on the reference, with the friction holding the cable spring.

3.2 Step 2: reference and commands

The reference of Section 1 gives the position command of each axis (relative to the start) and its rate, which is the input of the velocity feedforward. The elevation of the plant is set to the start of the reference, since the inertia of the azimuth depends on it.

3.3 Step 3: gyro and observer in the continuous loop

`GyroObserverLoop` (module `lucy_cl.disturbance_observer`) extends the closed loop with, per axis:

- **gyro** on the telescope: elevation $\omega_g = \dot{\theta}_p$ (telescope node); cross-elevation $\omega_g = \dot{\psi}_L \cos\theta_p$ (fork node), which the azimuth observer divides by the cosine of the encoder elevation (secant correction);
- **observer** (2.1), one state $\hat{\omega}$, with J' the rigid inertia of the axis and K_t' the nominal torque constant. The true torque is $\kappa_i K_t i$ with $\kappa_i = 0.96$ and 1.07: the observer works with a realistic 4–7 % error on K_t (Hilkert's sensitivity analysis [1, sec. 4]);
- **feedforward** $i_{ff} = -H_n \hat{T}_d / K_t'$ added to the current command after the velocity PI (the excitation input of the drive model), with H_n an optional chain of biquads.

3.4 Step 4: stability and the notch on the structural mode

- **Method.** The linearised loop (central differences of the time-domain model about rest, the Dahl elements becoming springs σ_0) is checked through its eigenvalues for each observer bandwidth $f_o = K_o/(2\pi J')$, with the observer on one axis at a time (Table 2).
- **Azimuth:** stable up to 80 Hz.
- **Elevation, no notch:** unstable from 8 Hz. The unstable mode is the first flexible mode of the elevation, about 216 Hz (telescope against motor): the gyro on the telescope and the motor torque are not collocated, so the observer closes a loop through the structure. This is the instability Hilkert finds when the observer bandwidth approaches the structural resonance [1, sec. 5].
- **The remedy is Hilkert's:** pass the estimate through a notch at the structural frequency [1, fig. 3, fig. 16]. With a notch at 210 Hz ($Q_z = 5$, $Q_p = 0.5$, -20 dB) the elevation is stable up to 40 Hz, also with the notch detuned by ±10 %.

Note. The first elevation mode measured in the October frequency responses is near 95 Hz, while the model places it at 210 Hz. In the real system the stability limits are lower and the notch has to be tuned on the measured mode (the elevation drive already has notches at 50 and 96 Hz on the velocity error).

Table 2: Stability of the linearised loop with the observer on one axis, against the observer bandwidth: stable, or the frequency of the unstable mode.

axis	notch	2 Hz	5 Hz	7 Hz	8 Hz	10 Hz	20 Hz	40 Hz	80 Hz
azimuth	no	stable	stable	stable	stable	stable	stable	stable	stable
elevation	yes	stable	stable	stable	stable	stable	stable	stable	**221 Hz**
elevation	no	stable	stable	stable	**217 Hz**	**216 Hz**	**215 Hz**	**215 Hz**	**218 Hz**

3.5 Step 5: metrics

- **Pointing error** [4, eq. 5.23]: the angle between the line of sight $\hat{n}(\psi_L, \theta_p)$ of the telescope and the direction $\hat{r}$ to the drone,

$$E = \left\| \hat{n}(\psi_L, \theta_p) \times \hat{r} \right\|, \qquad \hat{n} = [\cos\theta\cos\psi,\ \cos\theta\sin\psi,\ \sin\theta]^T, \tag{3.1}$$

  rms, 95th percentile and maximum over one orbit; also its rms times the distance (miss at the drone).
- **Jitter**: rms of the two components of the LOS error (cross-elevation $(\psi_{ref} - \psi_L)\cos\theta_{ref}$, elevation $\theta_{ref} - \theta_p$), high-passed above 1, 10 and 100 Hz (zero-phase second-order Butterworth). The three values are nested; the rms in a band follows as $\sqrt{J_{f_1}^2 - J_{f_2}^2}$.

3.6 Step 6: EtherCAT at 1 ms

`SampledObserverLoop` (module `lucy_cl.ethercat`) turns the loop into a sampled-data system with cycle $T = 1\,\text{ms}$ (Figure 5):

- **targets** (RxPDO, cyclic synchronous position): the master sends each cycle the target for the end of the cycle, and the drive interpolates linearly,

$$p(t) = p_{ref}(t_k) + \frac{p_{ref}(t_{k+1}) - p_{ref}(t_k)}{T}(t - t_k), \qquad v_{ff} = \frac{p_{ref}(t_{k+1}) - p_{ref}(t_k)}{T}, \quad t \in [t_k, t_{k+1}); \tag{3.2}$$

  the alternative of a target held over the cycle (zero-order hold) is also studied;

- **actual currents** (TxPDO): sampled at t_k and read by the master;

- **observer** in the master, discretised exactly for inputs held over the cycle, with $a = K_o/J'$:

$$\hat{T}_{d,k} = K_o(\omega_{g,k} - \hat{\omega}_k), \qquad \hat{\omega}_{k+1} = e^{-aT}\hat{\omega}_k + \big(1 - e^{-aT}\big)\Big(\omega_{g,k} + \frac{K_t' i_k}{J'a}\Big); \tag{3.3}$$

- **notch** in the master: bilinear transform prewarped at 210 Hz (below the 500 Hz Nyquist frequency);

- **feedforward current** (RxPDO): computed from the data of cycle k, applied by the drive from t_{k+1} and held for one cycle (one cycle of delay plus the hold).

The loops inside the drive (position, velocity, current) stay continuous. Between two cycles the plant, motors, drives and gyro dynamics are integrated by one Radau solver kept across the cycles (step size, Jacobian and factorisation are reused; the derivative is refreshed at each cycle start because the held inputs jump there).

3.7 Step 7: the MEMS gyro (ADIS16507)

`MemsGyro` (module `lucy_cl.gyro`) models one axis of the gyro of an ADIS16507-1, read over SPI (Table 3):

$$\omega_g(t_k) = \Big[H(s)\big(\omega + n_w + b\big)\Big]_{t=\lfloor (t_k - \tau_l) f_{odr} \rfloor / f_{odr}}, \qquad H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \tag{3.4}$$

- **dynamics:** second order with the $-3\,\text{dB}$ bandwidth of the data sheet (480 Hz, $\zeta = 0.707$), states of the continuous model;

- **digital output:** registers updated at 2000 SPS, read by the master $\tau_l = 0.5\,\text{ms}$ after the sample (data ready and SPI burst read); the latency is a first-order Padé delay in the continuous part;

- **noise from the Allan deviation** [6]:
  - angle random walk $N = 0.13\,\text{deg}/\sqrt{\text{h}} = 3.78{\cdot}10^{-5}\,\text{rad}/\sqrt{\text{s}}$: white rate noise, Allan deviation $N/\sqrt{\tau}$;
  - bias instability $B = 2.3\,\text{deg/h}$: first-order Gauss–Markov bias whose Allan deviation peaks at B at the time of the minimum, $\tau_B \approx 1.89 T_c$, with standard deviation $B/0.437$;
  - the data sheet gives neither the time of the minimum (taken as 50 s) nor a rate random walk (taken as zero). Over one orbit of 31 s these terms matter little: the observer differentiates the gyro, and the white noise dominates.

The noise is generated on a 16 kHz grid, filtered by the sensor dynamics and read at the cycle times; the same realisation is used in all cases. Check (Figure 6): the Allan deviation of a one-hour noise record follows the model and the data-sheet points, and the output noise is 0.069 deg/s rms against 0.068 deg/s of the data sheet.

Table 3: Gyro model: ADIS16507-1, x and y axes [5].

quantity	value	source
$-3\,\text{dB}$ bandwidth	480 Hz, second order, $\zeta = 0.707$	data sheet (ADIS16507-1/-2, x, y)
output data rate	2000 SPS	data sheet
latency to the master	0.5 ms (one output period)	assumption (data ready + SPI burst)
angle random walk	$0.13\,\text{deg}/\sqrt{\text{h}}$	data sheet (ADIS16507-1, x, y)
in-run bias stability	2.3 deg/h	data sheet (ADIS16507-1)
time of the Allan minimum	50 s	assumption
rate random walk, turn-on bias	0	not modelled
output noise, no filtering	0.068 deg/s rms (model 0.069)	data sheet

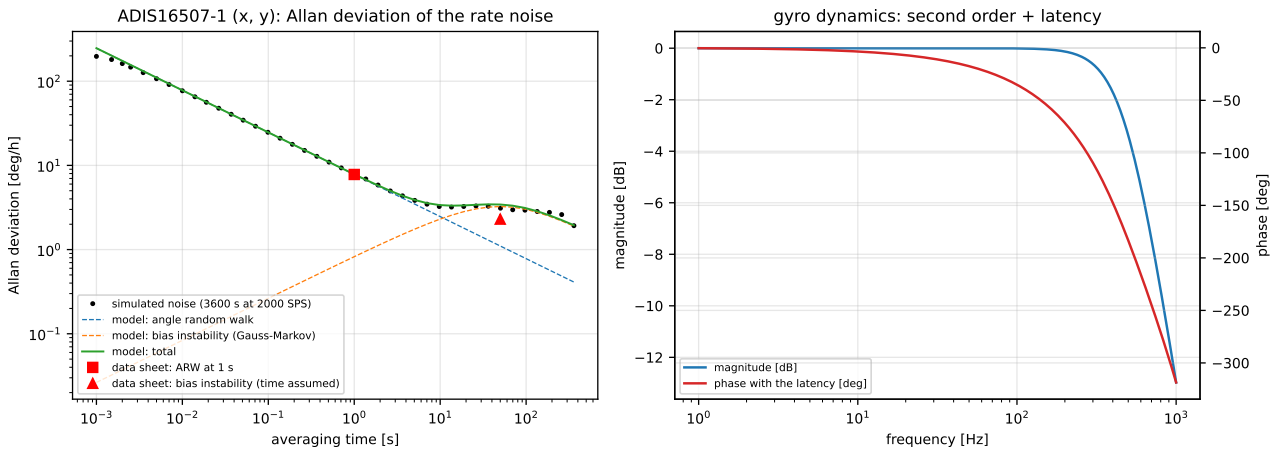


Figure 6: Gyro model. Left: Allan deviation [deg/h] against the averaging time [s]: simulated noise (one hour at 2000 SPS), model terms, data-sheet points. Right: magnitude [dB] and phase [deg] of the second-order dynamics with the latency.

4 Results with the continuous loop and an ideal gyro

Table 4: Continuous loop, ideal gyro: pointing error [mdeg], jitter [μrad], rms of the LOS error components [mdeg]; one orbit.

case	E rms	E max	jitter >1 Hz	jitter >10 Hz	cross-el. rms	elev. rms
current: loops on the encoders	124.11	183.7	99.4	1.89	123.82	8.48
encoders + gyro observer 5 Hz	117.66	182.6	30.5	2.03	117.51	6.03
encoders + gyro observer 10 Hz	117.66	182.6	7.93	0.586	117.50	6.04
encoders + gyro observer 20 Hz	117.66	182.6	4.42	0.0101	117.51	6.04
encoders + gyro observer 40 Hz	117.67	182.6	3.54	0.00888	117.51	6.04
current + velocity feedforward	26.61	130.9	99.4	1.89	26.22	4.59
current + velocity feedforward + gyro observer 20 Hz	2.70	7.8	4.42	0.0101	2.70	0.05
as above, gyro bandwidth 100 Hz	2.70	7.8	4.41	0.00957	2.70	0.05
current + velocity feedforward, alternative azimuth friction	11.79	41.7	3.61	0.00275	10.87	4.59
current + velocity feedforward + gyro observer 20 Hz, alternative azimuth friction	1.68	3.0	0.407	0.00132	1.68	0.05

- **Without the velocity feedforward** the pointing error is the following error of the P position loop, $\dot{\psi}/K_{pp}$, up to 182 mdeg in azimuth. The observer removes the part due to the disturbances (124 to 118 mdeg) but not this one: as Hilkert shows, in the ideal case the observer leaves the command response unchanged (eq. (2.2), [1, eq. 5]).
- **With the velocity feedforward** the disturbances dominate and the observer reduces the pointing error from 26.6 to 2.7 mdeg rms and the jitter above 1 Hz from 99 to 4.4 μrad.
- **The residual of 2.7 mdeg** is the twist of the azimuth shaft between the motor (where the encoder is) and the fork: the friction torque, cancelled by the motor, still passes through it ($1/k_{\psi,2} = 4.4$ mdeg per N m). The encoder follows its command at 0.54 mdeg rms while the LOS differs from it by 2.75 mdeg rms; a loop on the encoders cannot see it.
- **The disturbance estimate** follows the disturbance of the model (Figure 8); the small offset in elevation is the 7 % error of K_t'.
- **The linearised rejection** (Figure 9, Section 4.1) improves below the loop bandwidth by one to two orders of magnitude, as eq. (2.2) predicts; in azimuth it levels off at the compliance of the shaft.

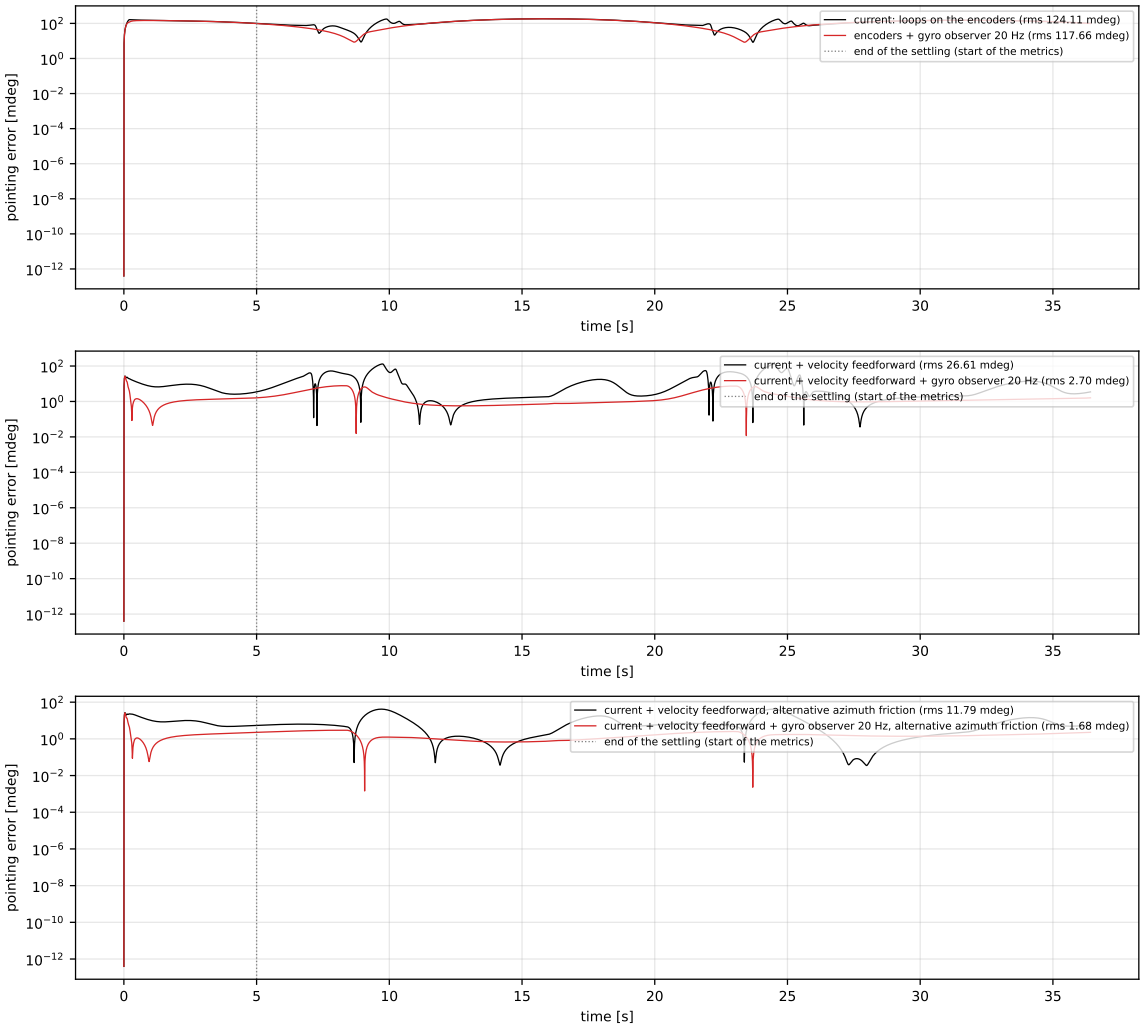


Figure 7: Continuous loop, ideal gyro: pointing error [mdeg, log scale] against time [s], without (top) and with (middle) the velocity feedforward, and with an alternative azimuth friction (bottom).

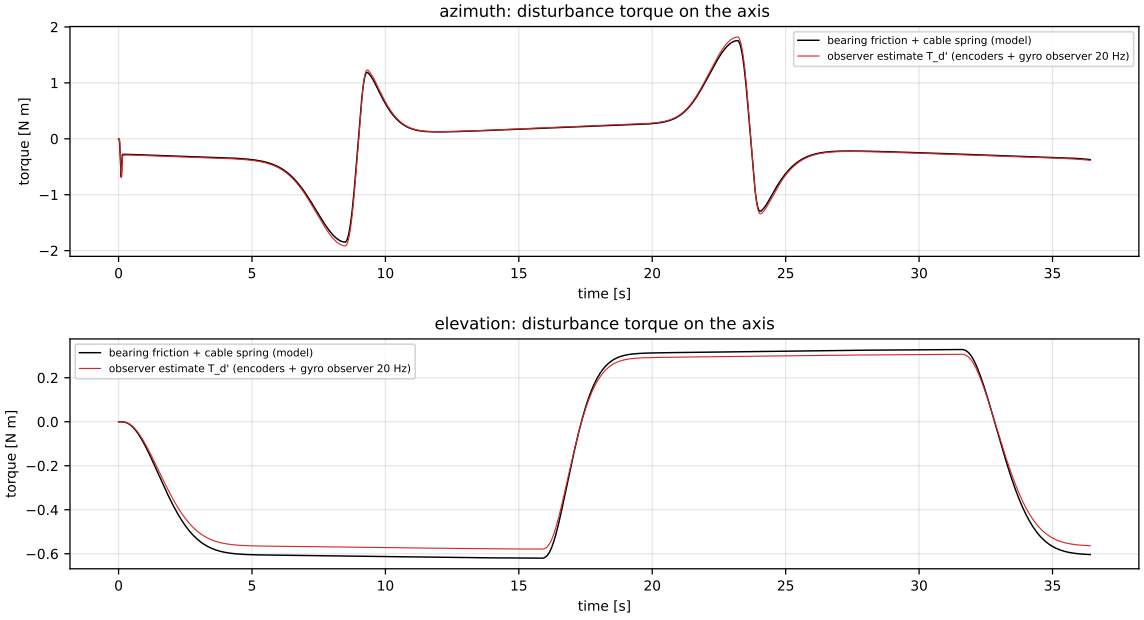


Figure 8: Disturbance torque on each axis [N m] against time [s]: bearing friction plus cable spring of the model, and the observer estimate $\hat{T}_d$ (20 Hz).

4.1 Disturbance rejection: definition and reading of the transfer function

Definition. The time-domain model of the closed loop (plant H6, motors, drives and, when present, gyro and observer) is linearised about rest at the start of the reference, by central differences, as $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}_d\, T_b$, $\theta_{LOS} = \mathbf{c}\,\mathbf{x}$, where:

- T_b is a torque applied to the *bearing node* of the axis (fork in azimuth, hub in elevation), the node where the friction and the cable spring act;
- $\mathbf{b}_d = [\mathbf{0};\ \mathbf{M}^{-1}(\mathbf{q}_0)\,\mathbf{e}_b;\ \mathbf{0}]$ puts it into the accelerations of the plant;
- $\mathbf{c}$ selects the line-of-sight coordinate (ψ_L, θ_p).

Figure 9 plots the magnitude of

$$G_d(j\omega) = \frac{\theta_{LOS}(j\omega)}{T_b(j\omega)} = \mathbf{c}\,(j\omega\mathbf{I} - \mathbf{A})^{-1}\mathbf{b}_d \qquad [\mathrm{rad}/(\mathrm{N\,m})]. \tag{4.1}$$

Why "sliding friction". Linearised at rest, the Dahl element of the bearing, $\dot{T}_f = \sigma_0\big(v - |v|\,T_f/g(v)\big)$, becomes a spring of stiffness σ_0 (pre-sliding): it describes micro-motions of a bearing that sticks, not of one that slides. While tracking, the bearing slides and its torque is set by the speed (Coulomb and Stribeck levels): towards the axis it acts as an external torque. The linearisation therefore sets $\sigma_0 \to 0$, so that the element leaves $\mathbf{A}$, and the friction becomes the input itself: $T_b = -T_f + T_{cable}$. The cable stiffness (1.7 N m/rad) stays in $\mathbf{A}$ and is negligible. Any other torque on the bearing (imbalance, wind, cable reaction) enters through the same function.

Meaning. By superposition:

- a sinusoidal bearing torque of amplitude T at frequency f moves the LOS by $|G_d(f)|\,T$;
- a random torque of spectral density $S_T(f)$ gives a LOS spectral density $|G_d|^2 S_T$ and a variance $\sigma_\theta^2 = \int |G_d|^2 S_T\, df$.

$|G_d|$ is thus the gain from the disturbance spectrum to the pointing and jitter spectrum: the lower, the better.

Rigid-body model. Below the first flexible mode, with an ideal current loop, the axis is an inertia J in the loop of the drive:

$$G_d(s) \simeq \frac{1}{Js^2 + \kappa_i K_t\, H_{fwd}(s)\, C_v(s)\big(C_p(s) + s\, H_{fb}(s)\big)} + c_{e\ell}, \tag{4.2}$$

with the transfer functions of the drive (Figure 2), and $c_{e\ell}$ the static compliance between the encoder and the LOS discussed below. Without filters the controller is a PID on the error, $k_D = K_{pv}$, $k_P = K_{pv}(K_{pp} + \omega_i)$, $k_I = K_{pv}K_{pp}\omega_i$ ($\omega_i = 2\pi K_{iv}$), and

$$G_d(s) \simeq \frac{s}{Js^3 + \kappa_i K_t\,(k_D s^2 + k_P s + k_I)}. \tag{4.3}$$

With the observer (rigid axis, exact parameters), Hilkert's eq. (6) [1] multiplies the rate response, and therefore the position response, by $s/(s+\omega_o)$: $G_{d,obs}(s) \simeq G_d(s)\, s/(s+\omega_o) + c_{e\ell}$. The dashed lines of Figure 9 are eq. (4.2) with the actual filters of the drive; they follow the full model up to the flexible modes.

Asymptotes.

- **Low frequency, current structure:** $|G_d| \simeq \omega/(\kappa_i K_t k_I)$, rising at +20 dB/decade. The velocity integrator rejects a constant torque completely at steady state. With $\kappa_i K_t = 0.249$ and 0.276 N m/A and $k_I = 4329$ and 6599 A/(rad s), this gives 583 and 345 μrad/(N m) at 0.1 Hz in azimuth and elevation, the same as the full model (Table 5).
- **Low frequency, with the observer:** $|G_d| \simeq \omega^2/(\kappa_i K_t k_I\, \omega_o)$, rising at +40 dB/decade and lower by ω/ω_o. In elevation the ratio to the current structure follows $|s/(s+\omega_o)|$ (Table 5, second number against third).
- **High frequency:** $|G_d| \to 1/(J\omega^2)$, −40 dB/decade, the same for all curves. Above the bandwidths of the loop and of the observer no controller acts and the axis responds as a free inertia (1.17 and 2.81 μrad/(N m) at 100 Hz). The observer cannot help there, as eq. (2.2) shows.

- **Maximum** at the rigid closed-loop poles:
  - azimuth: 1353 μrad/(N m) at 0.6 Hz (real poles at 0.26, 1.5 and 5.2 Hz);
  - elevation: 5704 μrad/(N m) at 2.8 Hz, where the current gains place a pair of poles with a damping ratio of only 0.29: a torque near 3 Hz is amplified.

Features.

- **Azimuth:**
  - the hump at 10–14 Hz and the steep fall after it come from the forward biquad of the azimuth drive (pole pair at 14 Hz, $Q = 3$, on the velocity error);
  - the dip at 65 Hz comes from the forward resonator at 65 Hz (+10 dB), which raises the loop gain there.
- **Elevation:**
  - the resonance and antiresonance at 210–230 Hz are the first flexible mode (telescope against motor) seen at the LOS;
  - the notches at 50 and 96 Hz lower the loop gain locally.

Compliance floor in azimuth. A bearing torque balanced by the motor twists the shaft between the motor (which carries the encoder) and the fork (the LOS). With the encoder held by its loop, the LOS moves by $c_{e\ell} = 76.4\,\mu$rad/(N m), computed from the static stiffness of the model ($= 1/k_{\psi,2}$).

- The observer cancels the acceleration of the axis but not this static twist, which the gyro does not see as a rate. At low frequency all its curves therefore tend to $c_{e\ell}$ (67–75 μrad/(N m) at 0.1 Hz).
- The current structure reaches the same floor only near 0.01 Hz, where $\omega/(\kappa_i K_t k_I)$ falls to $c_{e\ell}$.
- In elevation the encoder and the LOS lie on the same side of the torque path (hub to motor), so $c_{e\ell} = 0.0$.

Check against the simulations. At an azimuth reversal the friction swings from about +1.8 to −1.8 N m in 0.5–1 s, with content near 1 Hz.

- Half the swing times $|G_d(1\,\text{Hz})|$ gives $1.8 \times 1268\,\mu$rad ≈ 2.3 mrad $= 131$ mdeg for the current structure, and $1.8 \times 124\,\mu$rad ≈ 13 mdeg with the observer at 20 Hz.
- The time-domain maxima are 131 and 7.8 mdeg (Table 9).

Validity.

- The function is linear and assumes a sliding bearing. Where the bearing sticks (reversals, the elevation at 0.06 deg/s) the friction is not an external torque but the stiff pre-sliding spring; there only the time-domain simulations hold.
- The gyro noise is not in G_d; it is treated in Section 5.

Table 5: $|G_d|$ [μrad/(N m)] at selected frequencies, full linearised model; in brackets: ratio to the current structure; ratio predicted by Hilkert, $|s/(s+\omega_o)|$.

axis	f [Hz]	current	observer 5 Hz	observer 10 Hz	observer 20 Hz	observer 40 Hz
azimuth	0.1	580	66.9 (0.12; 0.02)	71.6 (0.12; 0.01)	74 (0.13; 0.005)	75.2 (0.13; 0.0025)
	1	1.27e+03	293 (0.23; 0.2)	178 (0.14; 0.1)	124 (0.098; 0.05)	101 (0.08; 0.025)
	3	664	410 (0.62; 0.51)	268 (0.4; 0.29)	181 (0.27; 0.15)	136 (0.21; 0.075)
	10	218	228 (1; 0.89)	212 (0.97; 0.71)	179 (0.82; 0.45)	150 (0.69; 0.24)
	30	13.5	12.7 (0.94; 0.99)	11.7 (0.87; 0.95)	9.29 (0.69; 0.83)	5.4 (0.4; 0.6)
elevation	0.1	343	7.17 (0.021; 0.02)	3.74 (0.011; 0.01)	2.03 (0.0059; 0.005)	1.17 (0.0034; 0.0025)
	1	2.56e+03	531 (0.21; 0.2)	281 (0.11; 0.1)	153 (0.06; 0.05)	88.6 (0.035; 0.025)
	3	5.62e+03	2.93e+03 (0.52; 0.51)	1.75e+03 (0.31; 0.29)	1e+03 (0.18; 0.15)	589 (0.1; 0.075)
	10	315	288 (0.92; 0.89)	232 (0.74; 0.71)	156 (0.5; 0.45)	96.9 (0.31; 0.24)
	30	33.1	34.1 (1; 0.99)	34 (1; 0.95)	31.7 (0.96; 0.83)	25.4 (0.77; 0.6)

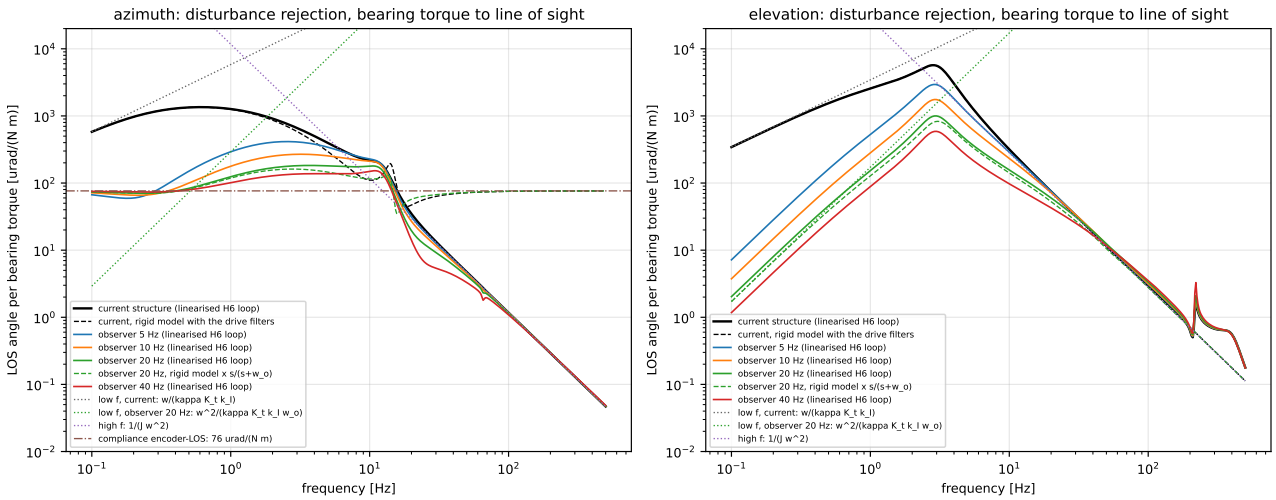


Figure 9: Disturbance rejection $|G_d|$, eq. (4.1): LOS angle per N m of torque on the bearing [μrad/(N m)] against frequency [Hz]. Solid lines: full linearised loops (sliding friction), current structure and observer bandwidths. Dashed lines: rigid-body model, eq. (4.2), with the filters of the drive, for the current structure and the observer at 20 Hz. Dotted lines: asymptotes $\omega/(\kappa_i K_t k_I)$, $\omega^2/(\kappa_i K_t k_I \omega_o)$ and $1/(J\omega^2)$. Dash-dotted line: compliance floor between the encoder and the LOS (azimuth).

5 Focus: pointing error and jitter, current structure against the observer with the noisy gyro

Constraints of Section 3 (EtherCAT 1 ms, ADIS16507 read over SPI), velocity feedforward on, observer at 20 Hz with the notch in elevation; the observer with a noise-free gyro (same EtherCAT and gyro dynamics) separates the effect of the noise.

Table 6: Pointing error [mdeg], miss at the drone [mm] and jitter by band [μrad]; one orbit.

	pointing error				jitter [μrad]			
case	rms	95 %	max	miss [mm]	>1 Hz	1–10 Hz	10–100 Hz	>100 Hz
current (encoders)	26.62	55.44	131.0	46.4	98.9	98.8	1.88	$4.63 \cdot 10^{-4}$
observer 20 Hz, ideal gyro	2.70	6.99	7.7	4.7	5.04	5.04	0.0112	$3.14 \cdot 10^{-6}$
observer 20 Hz, ADIS16507	2.72	6.92	8.0	4.8	9.65	9.29	2.61	0.142

Table 7: The same in degrees.

case	E rms [deg]	E max [deg]	jitter >1 Hz	1–10 Hz	10–100 Hz	>100 Hz [deg]
current (encoders)	0.027	0.13	$5.7 \cdot 10^{-3}$	$5.7 \cdot 10^{-3}$	$1.1 \cdot 10^{-4}$	$2.7 \cdot 10^{-8}$
observer 20 Hz, ideal gyro	$2.7 \cdot 10^{-3}$	$7.7 \cdot 10^{-3}$	$2.9 \cdot 10^{-4}$	$2.9 \cdot 10^{-4}$	$6.4 \cdot 10^{-7}$	$1.8 \cdot 10^{-10}$
observer 20 Hz, ADIS16507	$2.7 \cdot 10^{-3}$	$8.0 \cdot 10^{-3}$	$5.5 \cdot 10^{-4}$	$5.3 \cdot 10^{-4}$	$1.5 \cdot 10^{-4}$	$8.1 \cdot 10^{-6}$

5.1 Pointing error

- The observer lowers the pointing error from 26.6 to 2.72 mdeg rms (10 times) and from 131 to 8.0 mdeg at the peak (16 times); at 100 m this is a miss of a few millimetres. The peaks of the current structure are the velocity reversals of the azimuth, where the friction passes through its Stribeck peak (Figure 10).
- The gyro noise does not change the pointing error (2.70 against 2.72 mdeg): it is a zero-mean, mostly high-frequency error, while the pointing error is dominated by the shaft twist and by the low-frequency residual of the friction.
- EtherCAT at 1 ms is invisible at this level: with the targets held instead of interpolated the pointing error is 2.90 mdeg with the observer and 26.7 mdeg without.
- All cases are well inside the threshold of 0.2 deg recalled by Attorrese.

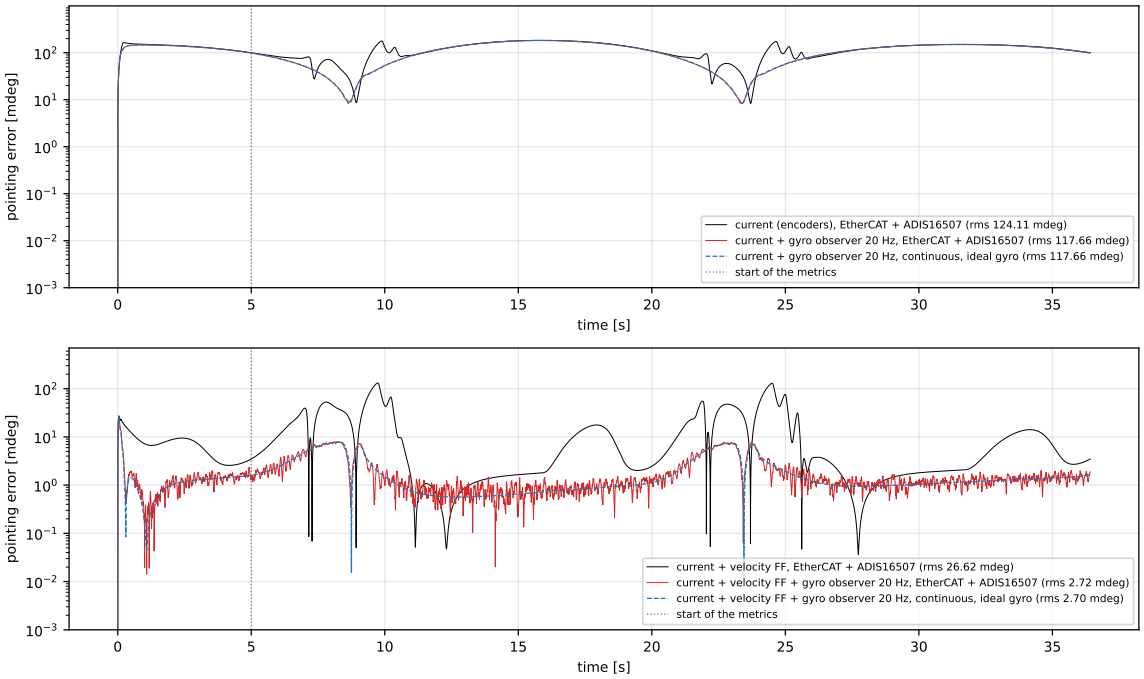


Figure 10: EtherCAT and ADIS16507: pointing error [mdeg, log scale] against time [s], current structure and observer, without (top) and with (bottom) the velocity feedforward; dashed: the observer in the continuous loop with an ideal gyro.

5.2 Jitter

Figure 11 splits the jitter by band and Figure 12 gives the rms of the LOS error above each frequency.

- **1–10 Hz, the band that dominates the jitter:** 99 μrad with the current structure, 9.3 μrad with the observer. Here sit the friction transients at the azimuth reversals, and here the observer acts, below its bandwidth. The jitter above 1 Hz therefore falls 10 times, from 99 to 9.7 μrad.

- **The gyro noise** adds about 8.2 μrad to the jitter above 1 Hz (5.0 μrad with an ideal gyro). Below its bandwidth the observer makes the LOS follow the inertial rate measured by the gyro, and therefore also the integral of its white noise (angle random walk) between the bandwidth of the position loop ($f_1 \approx 1$ Hz) and that of the observer (f_o):

$$\sigma_{LOS} \approx \frac{N}{2\pi}\sqrt{\frac{1}{f_1} - \frac{1}{f_o}} \approx 5.9\ \mu\text{rad}, \tag{5.1}$$

  consistent with the simulation. This floor scales with the angle random walk of the gyro.

- **10–100 Hz:** 2.61 μrad with the observer against 1.88 μrad, slightly worse. With an ideal gyro this band would almost vanish (0.011 μrad above 10 Hz); with the ADIS16507 it is the gyro noise.

- **Above 100 Hz:** 0.14 μrad against $4.6{\cdot}10^{-4}$ μrad, more than two orders of magnitude higher but negligible in absolute terms (less than 0.1 % of the total). The current structure is that clean only because the model has no other noise source (encoder quantisation and noise, current measurement, PWM): the comparison in this band is not a fair one.

- **Noise-equivalent current** (Figure 14): the feedforward current of the azimuth carries about 1.3 A rms of noise (2.86 A rms against 2.55 A with an ideal gyro), flat between the observer bandwidth and the gyro bandwidth, as Hilkert's eq. (10) predicts [1]. It heats the motor and uses part of the current margin, but does not reach the LOS, filtered by the inertia.

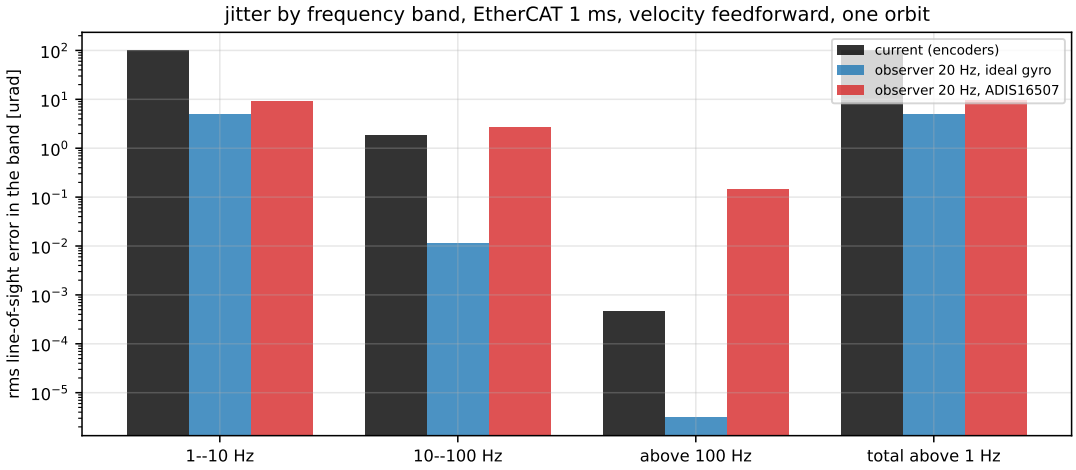


Figure 11: Jitter by frequency band [μrad, log scale]: rms of the LOS error in 1–10 Hz, 10–100 Hz, above 100 Hz and in total above 1 Hz; current structure, observer with an ideal gyro, observer with the ADIS16507.

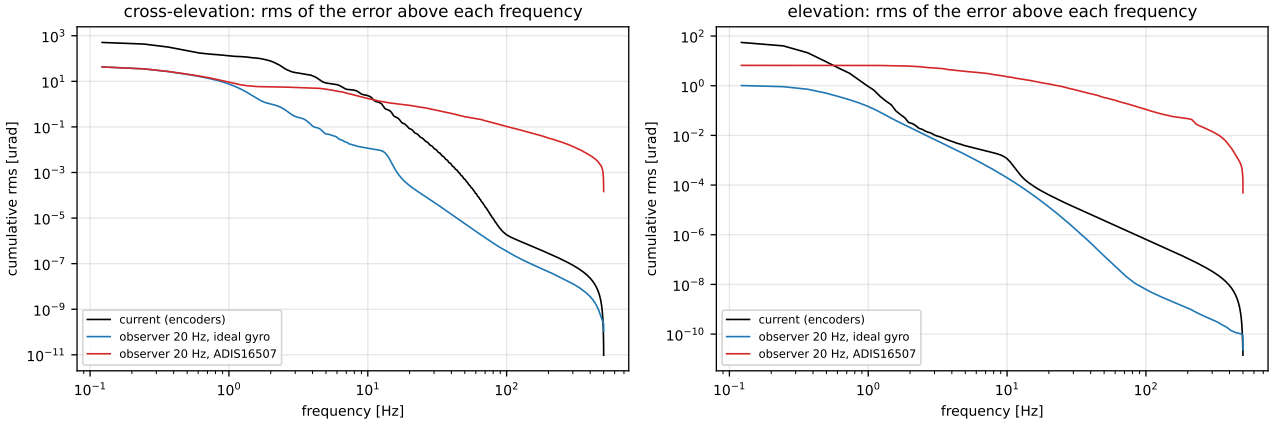


Figure 12: Rms of the LOS error above each frequency [μrad] against frequency [Hz], cross-elevation and elevation: where the jitter comes from.

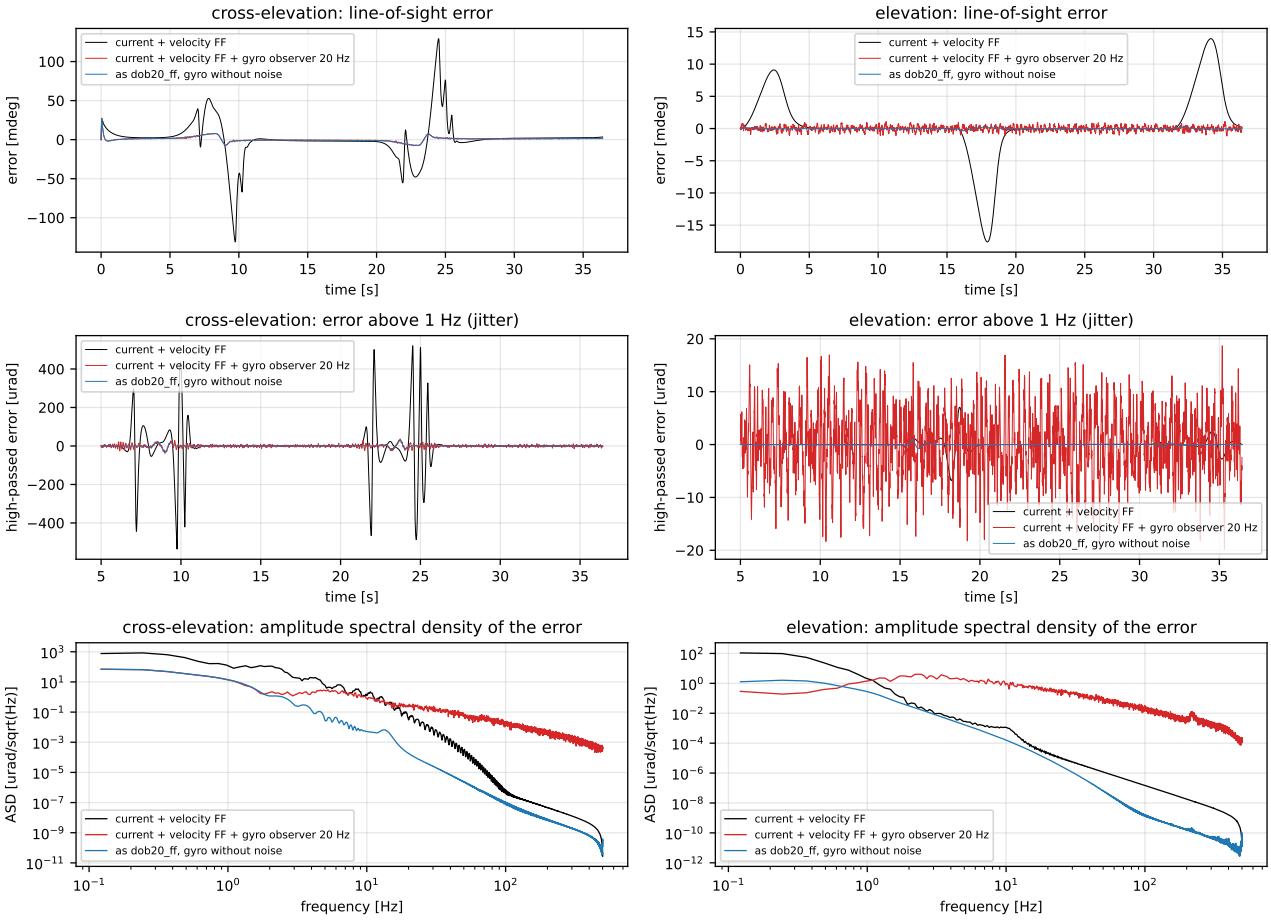


Figure 13: EtherCAT and ADIS16507, velocity feedforward: LOS error components [mdeg] against time [s] (top), error above 1 Hz [μrad] (middle) and amplitude spectral density [$\mu\text{rad}/\sqrt{\text{Hz}}$] (bottom); current structure, observer with the noisy gyro, observer with an ideal gyro.

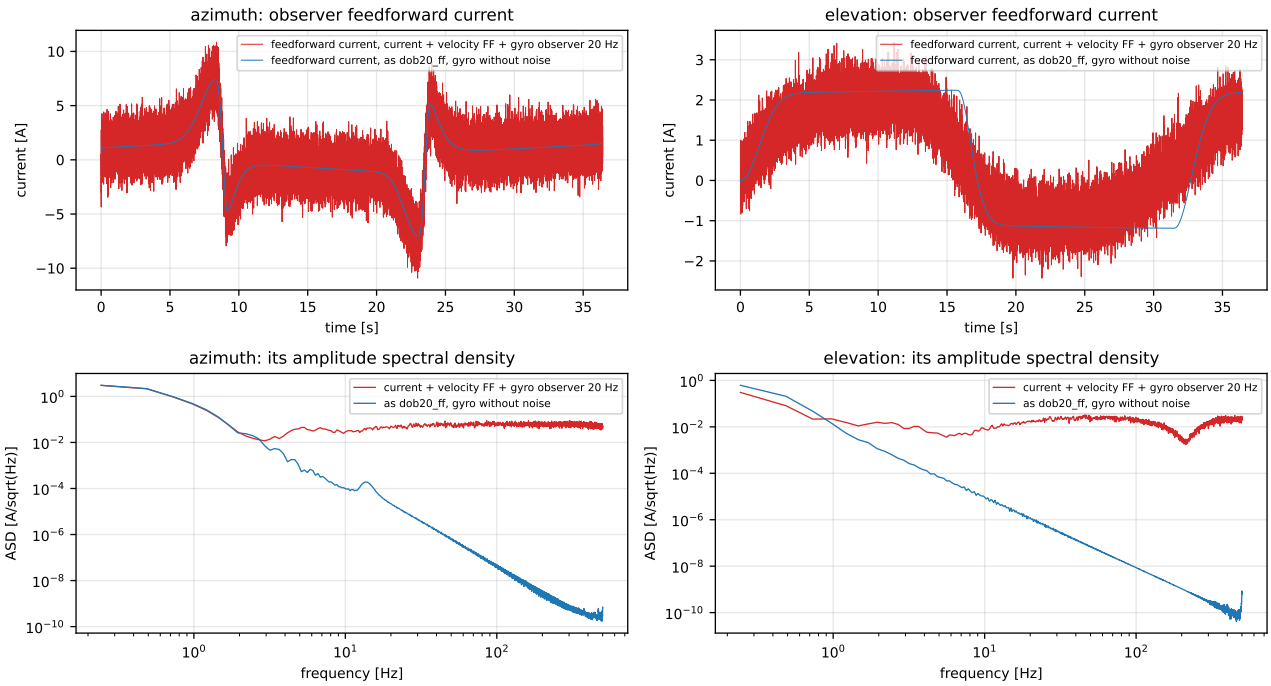


Figure 14: Feedforward current of the observer [A] against time [s] (top) and its amplitude spectral density [$\text{A}/\sqrt{\text{Hz}}$] (bottom), noisy and ideal gyro.

5.3 Required motor torque

Figure 15 and Table 8 give the motor torque $\tau_m = \kappa_i K_t\, i$ of the three cases of Table 6.

- **What the torque is made of.**
  - The inertial part of the reference is negligible: $J\ddot{\theta}_{ref}$ is at most 9.1 mN m in azimuth and 0.2 mN m in elevation.
  - The cable spring is almost constant (-0.04 and -0.15 N m).
  - The required torque is therefore the friction torque. In azimuth it is about ±0.2–0.4 N m while sliding, with peaks of ±1.84 N m at the velocity reversals (Stribeck peak $T_s = 1.84$ N m). In elevation it is close to the break-away level $T_s = 0.46$ N m plus the cable spring, with the sign set by the direction of the slow motion.
- **Current structure against the observer with an ideal gyro.** The rms and peak torques are the same (0.65 and 0.63 N m rms in azimuth, 0.46 and 0.46 in elevation): the observer does not ask for more torque, it delivers the same torque *earlier*.
  - At the azimuth reversal (middle of Figure 15) the observer reverses the torque as soon as the friction does.
  - The current structure keeps it for about another second, while its velocity integrator winds up through the error. That delay is the 131 mdeg peak of the pointing error.
  - In elevation the same holds at the reversal of the slow motion.
- **With the ADIS16507:**
  - the torque carries the gyro noise, differentiated by the observer: 0.33 N m rms above 10 Hz in azimuth and 0.11 in elevation, with a flat spectrum from the observer bandwidth to the gyro bandwidth (Figure 15, bottom; the elevation notch at 210 Hz is visible);
  - the azimuth peak grows to 2.69 N m (40 % of the peak torque of the motor, 6.74 N m);
  - the rms grows by about 10 % and the copper losses, proportional to i^2, by about 20 %.

  All values stay well inside the limits of the motor (continuous stall torque 2.18 N m).
- **Elevation, lower low-frequency torque with the noisy gyro** (0.36 against 0.46 N m). The noise current acts on the bearing as a *dither* [3, sec. 11.2.2]: the Dahl element, cycled by the high-frequency torque, carries a lower mean friction (mean $|T_f|$ 0.30 against 0.43 N m). This is a property of the friction model; a real bearing behaves qualitatively the same, but the size of the effect should not be taken at face value.

Table 8: Motor torque $\kappa_i K_t i$ over one orbit, EtherCAT and velocity feedforward: rms and peak [N m], rms of its parts below and above 10 Hz [N m], rms against the continuous stall torque and peak against the peak torque of the motor [%].

axis	case	rms	peak	rms < 10 Hz	rms > 10 Hz	rms/T_{cont}	peak/T_{peak}
azimuth	current (encoders)	0.646	1.84	0.645	0.012	30	27
	observer 20 Hz, ideal gyro	0.633	1.84	0.633	0.000	29	27
	observer 20 Hz, ADIS16507	0.709	2.69	0.627	0.330	33	40
elevation	current (encoders)	0.461	0.62	0.461	0.000	21	9
	observer 20 Hz, ideal gyro	0.460	0.62	0.460	0.000	21	9
	observer 20 Hz, ADIS16507	0.374	0.97	0.357	0.109	17	14

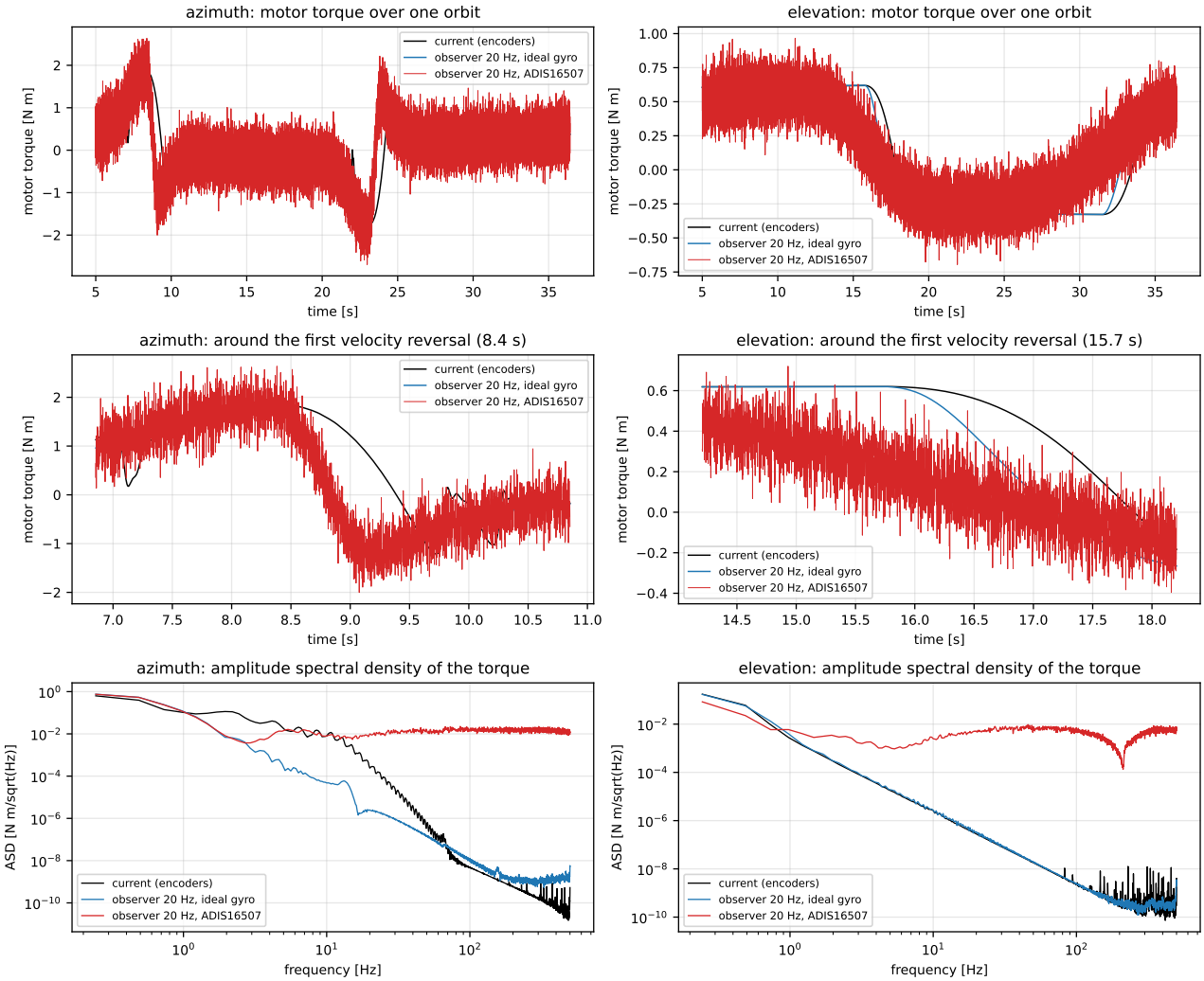


Figure 15: Motor torque [N m] of the current structure and of the observer at 20 Hz with an ideal and with the ADIS16507 gyro, EtherCAT and velocity feedforward: over one orbit against time [s] (top), around the first velocity reversal of each axis (8.4 and 15.7 s, middle), and amplitude spectral density [N m/$\sqrt{\text{Hz}}$] against frequency [Hz] (bottom).

5.4 Choice of the observer bandwidth

Figure 16 repeats the comparison for observer bandwidths of 5–40 Hz, with the noisy and the ideal gyro.

- The jitter above 1 Hz is 42.5, 18.4, 9.7 and 10.3 μrad at 5, 10, 20 and 40 Hz: below 20 Hz the friction transients are not rejected enough, above it the gyro noise grows faster than the rejection. 20 Hz is the best of the four.
- The jitter above 10 and 100 Hz and the noise current grow with the bandwidth, as Hilkert's figures 10–13 show [1].
- The pointing error does not improve beyond 10–20 Hz.

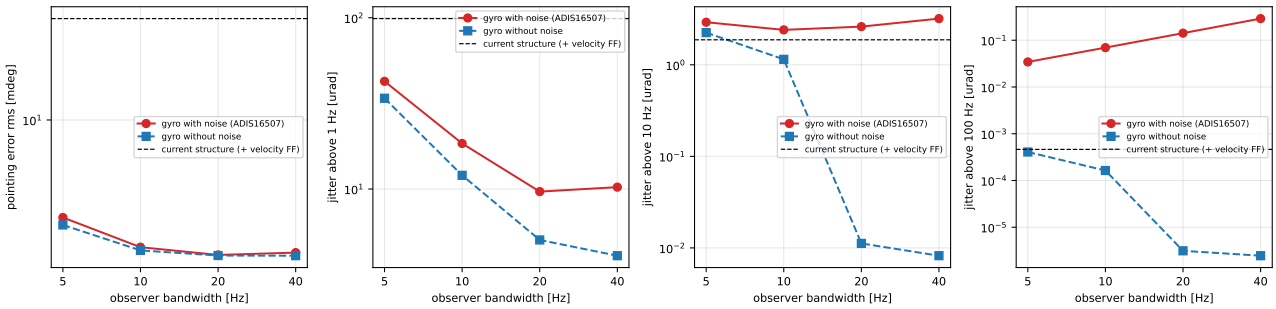


Figure 16: EtherCAT, velocity feedforward: pointing error rms [mdeg] and jitter above 1, 10 and 100 Hz [μrad] against the observer bandwidth [Hz], noisy and ideal gyro, and the current structure.

5.5 Overall table

Table 9: EtherCAT and ADIS16507, all cases: pointing error [mdeg], jitter [μrad], feedforward current rms azimuth / elevation [A]; one orbit.

case	E rms	E max	jitter >1 Hz	>10 Hz	>100 Hz	i_{ff} rms
current (encoders)	124.11	183.7	98.9	1.88	0.000463	0.00 / 0.00
current + gyro observer 20 Hz	117.66	183.0	9.66	2.63	0.142	2.86 / 1.35
current + velocity FF	26.62	131.0	98.9	1.88	0.000463	0.00 / 0.00
current + velocity FF + gyro observer 5 Hz	3.90	15.2	42.5	2.92	0.0344	2.47 / 1.63
current + velocity FF + gyro observer 10 Hz	2.93	10.1	18.4	2.41	0.0695	2.62 / 1.55
current + velocity FF + gyro observer 20 Hz	2.72	8.0	9.65	2.61	0.142	2.86 / 1.35
current + velocity FF + gyro observer 40 Hz	2.78	8.5	10.3	3.2	0.29	3.72 / 1.21
as dob5_ff, gyro without noise	3.63	15.0	33.9	2.25	0.000407	2.52 / 1.67
as dob10_ff, gyro without noise	2.84	9.8	12	1.15	0.000164	2.55 / 1.66
as dob20_ff, gyro without noise	2.70	7.7	5.04	0.0112	3.14e-06	2.55 / 1.66
as dob40_ff, gyro without noise	2.70	7.8	4.08	0.00822	2.46e-06	2.55 / 1.66
current + velocity FF, targets held (zoh)	26.70	131.2	98.9	1.88	0.000463	0.00 / 0.00
current + velocity FF + gyro observer 20 Hz, targets held (zoh)	2.90	8.0	9.65	2.61	0.142	2.86 / 1.35

6 Limits and next steps

Limits.

- The structural mode of the elevation in the model (210 Hz) is not the measured one (about 95 Hz): the stability limits and the notch must be revised on the real mode.
- The azimuth Stribeck peak ($T_s/T_c = 7.9$) is the least determined parameter of the identification. With the alternative friction of the identification ($T_s/T_c = 2.66$) the absolute values change, but the observer still gains 7–9 times (continuous study).
- Only the gyro is noisy: encoder quantisation and noise, current measurement noise and PWM are not modelled, so the high-frequency jitter of the current structure is optimistic.
- The drive loops are continuous; the data-sheet values of the gyro without a time of the Allan minimum and a rate random walk are completed by assumptions.

Next steps.

- A low pass on the observer output (or on both its inputs, not on the gyro alone, which would create a false disturbance during the accelerations) at 3–5 times the observer bandwidth: it halves the noise current and lowers the jitter above 100 Hz, but not the dominant 1–10 Hz part.
- A gyro with a lower angle random walk (the noise floor scales with it), or a lower observer bandwidth where the friction allows it.
- The twist of the azimuth shaft (2.7 mdeg) needs the LOS position: a loop on the integrated gyro or on an optical sensor, or the predicted twist $\hat{T}_d/k_{\psi,2}$ added to the position command.
- Encoder and current noise in the model, for a fair comparison of the high-frequency jitter.

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[1] J. M. Hilkert, B. Pautler, "A reduced-order disturbance observer applied to inertially stabilized line-of-sight control", *Proc. SPIE* 8052, Acquisition, Tracking, Pointing, and Laser Systems Technologies XXV, 80520H, 2011. doi:10.1117/12.884123.

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[5] Analog Devices, *ADIS16507 Precision, Miniature MEMS IMU*, data sheet Rev. A.

[6] IEEE Std 952-1997, *IEEE Standard Specification Format Guide and Test Procedure for Single-Axis Interferometric Fiber Optic Gyros* (Annex C: Allan variance noise terms).